

MASON JEWETT, MI, USA

WMO: 725417

Lat: 42.566N

Lon: 84.433W

Elev: 281

StdP: 97.99

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 54822

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.9	-15.0	-22.1	0.5	-16.9	-19.0	0.7	-13.8	9.5	-3.9	8.5	-3.8	1.4	220	0.487

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	32.0	22.3	30.2	21.6	28.6	20.8	24.1	28.9	23.1	27.9	22.2	26.7	3.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	17.8	26.7	21.9	17.2	26.0	21.0	16.2	25.0	73.6	29.1	70.1	27.7	66.7	26.4	35.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.7	7.6	6.9														
			DB	-22.8	34.2	3.3	3.0	-25.1	36.4	-27.1	38.2	-28.9	39.9	-31.3	42.1	
			WB	-23.0	26.4	3.3	2.5	-25.4	28.2	-27.3	29.6	-29.1	31.0	-31.5	32.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.3	-4.5	-3.5	1.9	8.7	14.9	19.9	21.7	20.7	17.1	10.7	4.5	-1.5
	DBStd	10.54	6.24	6.04	6.15	5.14	4.87	3.71	3.21	3.07	4.27	4.96	5.06	5.50
	HDD10.0	1777	449	379	262	84	12	0	0	0	2	53	178	359
	HDD18.3	3643	707	611	510	291	131	26	8	12	72	243	416	616
	CDD10.0	1513	1	1	11	45	162	296	363	333	216	73	12	1
	CDD18.3	338	0	0	1	2	23	72	112	86	37	6	0	0
	CDH23.3	2955	0	0	6	27	222	618	1031	696	313	43	0	0
	CDH26.7	844	0	0	1	3	54	184	331	183	83	7	0	0
Wind	WSAvg	2.7	3.2	3.2	3.2	3.5	2.8	2.3	2.0	1.8	2.0	2.6	3.0	3.1
Precipitation	PrecAvg	878	59	51	60	75	102	98	88	73	78	73	69	52
	PrecMax	1064	107	95	107	156	245	188	168	158	248	167	132	95
	PrecMin	685	19	21	19	9	49	51	23	12	18	10	32	20
	PrecStd	104	27	22	22	34	50	35	34	38	52	35	31	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.2	14.9	23.7	27.0	30.9	32.4	33.8	32.9	32.0	27.6	21.1	14.9
		MCWB	11.3	11.5	17.2	16.6	20.7	22.2	23.1	22.3	21.1	20.8	14.2	12.6
	2%	DB	9.1	11.3	17.7	23.0	27.9	31.0	32.1	30.6	29.0	24.0	17.1	11.3
		MCWB	7.1	9.0	12.6	14.5	19.0	21.7	23.1	22.1	20.8	17.2	12.0	9.4
	5%	DB	6.2	7.3	14.1	21.0	26.1	28.8	30.1	28.6	27.1	21.5	14.8	7.8
		MCWB	4.2	4.8	10.3	13.5	17.9	20.7	22.1	21.0	19.4	16.2	11.1	6.0
	10%	DB	3.6	4.0	11.3	17.8	23.2	27.2	28.0	27.3	24.9	18.9	12.4	5.3
		MCWB	2.3	2.0	7.9	11.8	16.5	19.7	20.9	20.4	18.6	14.9	9.1	3.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.1	12.7	17.8	18.5	22.3	24.1	25.7	24.9	23.7	21.6	15.4	12.9
		MCDB	13.0	14.0	23.1	24.4	27.7	29.5	30.9	30.2	28.8	26.8	19.0	14.2
	2%	WB	7.8	8.9	13.9	16.3	20.6	22.9	24.4	23.5	22.3	19.3	13.2	9.7
		MCDB	8.5	10.9	16.7	20.1	25.8	28.3	29.3	27.8	26.5	22.3	16.2	11.3
	5%	WB	4.4	5.0	11.4	14.6	19.4	21.9	23.5	22.5	21.1	17.1	11.5	6.3
		MCDB	6.1	7.1	13.9	19.5	23.9	26.8	28.2	26.7	24.7	19.9	14.2	7.7
	10%	WB	2.4	2.3	8.0	12.5	18.0	20.9	22.4	21.6	19.8	15.2	9.3	3.6
		MCDB	3.7	4.2	10.5	17.2	22.1	25.2	26.8	25.5	23.6	18.5	11.9	5.1
Mean Daily Temperature Range	5% DB	MDBR	7.2	8.0	9.3	11.1	11.1	10.7	11.4	11.0	11.4	10.0	8.1	6.4
		MCDBR	9.0	10.3	13.5	14.6	13.2	12.4	12.8	12.3	13.3	13.3	11.6	9.3
		MCWBR	7.7	8.6	9.2	8.2	6.8	5.6	5.7	5.6	6.2	7.5	8.1	7.9
	5% WB	MCDBR	8.3	10.0	12.5	12.7	11.3	10.4	10.5	9.8	10.5	11.1	10.7	9.1
		MCWBR	7.5	8.9	9.5	8.6	6.9	5.6	5.5	5.3	5.8	7.8	8.6	8.2
Clear-Sky Solar Irradiance	taub	0.295	0.345	0.342	0.387	0.417	0.438	0.427	0.421	0.381	0.345	0.314	0.296	
	taud	2.370	2.190	2.359	2.253	2.220	2.197	2.235	2.247	2.345	2.443	2.477	2.414	
	Ebn at Noon	843	839	901	879	859	839	844	837	847	841	819	810	
	Edh at Noon	80	114	109	131	139	143	136	130	110	88	73	71	
All-Sky Solar Radiation	RadAvg	1.46	2.35	3.59	4.57	5.33	6.02	6.12	5.25	4.23	2.71	1.69	1.16	
	RadStd	0.13	0.30	0.36	0.42	0.49	0.38	0.28	0.24	0.39	0.26	0.22	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (63 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page